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Project type: NoSQL – Comparison between a Graph Database and a Relational DBMS

Dataset: European Soccer Database (Kaggle). The dataset contains information about soccer matches, players, teams, and leagues across several European countries. It includes approximately 300,000 match records, 11,000 players, and 300 teams, covering 11 leagues over 8 seasons. Source: <https://www.kaggle.com/datasets/hugomathien/soccer>

Description: The project aims to compare a relational approach (PostgreSQL) with a graph-based approach (Neo4j) for the analysis of soccer data. Both systems will be modeled starting from a shared conceptual schema, designed to support non-trivial queries involving relationships between players, teams, leagues, and countries. The comparison will focus on multiple dimensions: (i) complexity and expressiveness of the query languages (SQL vs. Cypher) on equivalent analytical tasks; (ii) performance on graph-traversal queries (e.g., finding players connected through shared teams); (iii) ease-of-use and suitability of each model for the given domain. The results will highlight the strengths and limitations of each approach in the context of sports data analysis.